Nishka Shrimali

Hyderabad, India   +91-6291764011   shrimalinishka30@gmail.com   GitHub   LinkedIn

SUMMARY

Software Engineer with experience building production-grade **full-stack applications**, **REST APIs**, and scalable backend systems using Angular, ReactJS, .NET/C#, Node.js, and SQL/PostgreSQL. Skilled in **microservices**, secure authentication (**SSO/OAuth2/JWT**), **Azure** cloud deployment, and **AI-powered workflows**, with domain experience in **healthcare/EMR systems** and **FHIR/HL7** standards.

SKILLS

- **Languages:** JavaScript, TypeScript, C#, Java, SQL, Python
- **Frontend:** Angular, ReactJS, Next.js, Redux, jQuery, HTML5, CSS3, TailwindCSS, TanStack Grid, Flutter (Mobile)
- **Backend & APIs:** Node.js, Express.js, .NET Core / Web APIs, C#, FastAPI, REST APIs, Microservices, OAuth2, JWT, Redis
- **Databases & Data Engineering:** SQL (query optimization), PostgreSQL, MySQL, MongoDB, Firebase, Pinecone, Dimensional Modeling, Databricks
- **AI & LLM Systems:** LangChain, AutoGen, Semantic Kernel, MAF, MCP/FastMCP, RAG Pipelines, LLM Integration, Multi-Agent Systems, LightGBM, SHAP
- **Cloud & Dev Tools:** Azure, AWS, Docker, Kubernetes, OpenShift, CI/CD (Azure DevOps / GitHub Actions), Git, GitHub, Postman
- **Domain:** Healthcare / EMR Systems, FHIR, HL7

EXPERIENCE

Prolifics Corporation Ltd. — Hyderabad, India
Associate Software Engineer — Jul 2025 – Present

- Developed and enhanced **ITMS**, a production-grade logistics and transportation management platform for **Inter-Metro Freight**, a U.S.-based freight carrier operating across the continental United States, using ReactJS, C#, .NET Web APIs, and SQL, improving UI responsiveness and reducing page load times by 40–50% while implementing secure **Single Sign-On (SSO)** authentication.
- Optimized dimensional data models and SQL reporting workflows, reducing query execution time from nearly 200ms to 70–80ms across high-volume transportation operations.
- Built a real-time communication and notification system for **Inter-Metro Freight** using **Node.js**, **Twilio**, and webhook-driven workflows, enabling secure calling and messaging between drivers, warehouses, and head office teams across 6+ U.S. locations.
- Built an internal developer platform for generating configurable AI **backend templates** with **pluggable LLM integrations, authentication modules, and reusable orchestration workflows**.
- Designed and integrated **LLM-powered workflows** including prompt orchestration, **RAG pipelines**, context-aware execution, tool-calling capabilities, and **multi-agent orchestration** for intelligent backend automation.
- Developed an **MCP-based orchestration framework** enabling AI agents to dynamically generate and execute tools from natural language prompts using modular tool registries and runtime execution pipelines.

Future Byte Innovations — Remote
Frontend Intern — Nov 2024 – Jun 2025

- Developed a full-stack **Lead Management CRM** platform using Next.js, Node.js, MongoDB, and Google Maps API for lead tracking, geolocation workflows, customer management, and real-time data processing.
- Designed and optimized **RESTful APIs**, backend business logic, and MongoDB query workflows, improving application performance by 40% and reducing data retrieval latency by 30%.
- Implemented role-based authentication, API integrations, form validation, and an **AI-powered chatbot** using **OpenAI APIs** to automate user interactions, streamline lead workflows, and ensure secure data access.

Skepsi.AI — Remote
SDE Intern — Jun 2024 – Aug 2024

- Developed a full-stack hotel management platform using ReactJS, Node.js, and MongoDB, building scalable backend services for **role-based authentication**, operational workflows, and real-time booking management.
- Designed **RESTful APIs** and backend modules for room allocation, customer management, and live booking status tracking, improving operational visibility and workflow efficiency for hotel staff.
- Optimized backend query handling, API response flows, and frontend state management, improving dashboard responsiveness by 40% across high-traffic operational workflows.

PROJECTS

Monitoring MCP Server AI Agent *Python, FastMCP, MCP, Azure OpenAI, YAML, SQL, Docker*

- **Architected and independently built a reusable MCP Toolkit** that enables developers to create and deploy MCP tools using only **Python implementations and YAML configurations**, eliminating the need to modify the core server.
- Engineered a **dynamic tool lifecycle** with automatic discovery, schema validation, **AST-based security scanning**, registration, and runtime execution, enabling new tools to be added as modular plugins.
- Built an **Azure OpenAI-powered Monitoring Agent** that intelligently selects and invokes MCP tools based on natural-language requests, executes monitoring workflows, queries operational data, and generates contextual responses.
- Implemented **secure database monitoring** with parameterized read-only SQL execution, authentication, audit logging, health checks, retries, and controlled tool execution to prevent unsafe operations.
- Exposed the toolkit through **Streamable HTTP MCP** interfaces and containerized the system with **Docker**, creating a reusable foundation for integrating AI agents with enterprise monitoring and operational workflows.
- Added automated testing and CI workflows to validate **tool discovery, security checks, registration, agent orchestration, and execution paths** across the platform.

Sentinel - Deployment Risk Analyzer *Python, LightGBM, SHAP, networkx, MCP, LLM*

- Built a **CLI tool that predicts whether a code change will break production**, scoring each deployment with a **LightGBM** model trained on bug-fixing history mined from the repository itself via the **SZZ algorithm**; outperformed a lines-changed baseline on a 4,800-commit codebase (ROC-AUC 0.74, PR-AUC 0.25) under a strict **time-based train/test split** to prevent data leakage.
- Engineered a dependency-graph **blast-radius analyzer** (**networkx**) that maps direct and transitive dependents of changed files across Python and Java, detecting circular imports and hub modules to surface the true downstream impact of a change.
- Added an **explainability layer** using **SHAP** for per-factor risk attribution and grounded natural-language explanations via an **LLM** (NVIDIA OpenAI-compatible API), architected so the model can *explain* the score but structurally cannot alter it.
- Shipped across three surfaces on a single versioned JSON contract - **CLI**, an **MCP server** callable by AI agents and editors, and a **GitHub Action** that gates pull requests - backed by **298 automated tests**.

CareNexus - Healthcare Risk Analytics Platform *HAPI FHIR, PostgreSQL, Synthea, HL7/FHIR*

- Built a full-stack **clinical risk-analytics platform** on a self-hosted **HAPI FHIR** server backed by **PostgreSQL**, ingesting synthetic patient records generated with **Synthea** to model realistic **EMR** data.
- Designed three clinical risk-scoring modules - **Social Determinants of Health (SDOH)**, **Clinical Severity**, and **Medication Adherence** - combined into a single **weighted overall patient risk score**.
- Authored **PostgreSQL** setup and transformation scripts mapping **FHIR resources** into query-optimized relational structures that power the scoring pipeline and analytics dashboards.
- Produced technical architecture documentation and iteratively tuned score distributions for senior architect and stakeholder review.

ACHIEVEMENTS

- **Hackathon Achievements:**
  Finalist, Hackfest'23 (IIT ISM Dhanbad)
  2nd Runner-up, HackONova (Adamas University)
- **Problem Solving:**
  Solved 500+ Data Structures & Algorithms problems across multiple platforms

EDUCATION

MCKV Institute of Engineering 2021–2025
B.Tech in Computer Science and Engineering CGPA: 9.6